Shivam Gupta

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EDUCATION

University of Texas at Austin Aug 2018 – Dec 2024
Ph.D. in Computer Science (Advisor: Eric Price)
Dissertation: Diffusion Models – Training, Sampling, and Reconstruction

University of Illinois at Urbana-Champaign Aug 2014 – May 2018
B.S. in Computer Science, Minor in Mathematics

INTERESTS

Large Language Models, Diffusion Models, Machine Learning, Statistics, related topics

EXPERIENCE

Google DeepMind *New York, NY*
Senior Research Scientist Feb 2025 – Present
- Post-training research and large-scale training for Gemini
- Worked on SFT & RL algorithms, data, and reward modeling – with a focus on agentic/coding and reasoning performance
- Worked on improving the behavior and alignment of Gemini models, especially for SWE tasks – reducing hallucination, cheating, and improving code quality
- Contributed to Gemini 3.7 Flash, and Gemini 4 Argon, as well as internal large models

Google DeepMind *San Francisco, CA*
Student Researcher May 2024 – Oct 2024
- Worked on data mixture optimization techniques for LLM post-training
- Scaled up training setup to $100+$ TPUs, and set up data and evaluation pipelines from scratch for experimentation

Massachusetts Institute of Technology *Cambridge, MA*
Visiting Student Jun 2023 – Aug 2023
- Worked on sharper rates for high-probability mean estimation. Resulted in paper at COLT 2024.

University of California, Berkeley *Berkeley, CA*
Visiting Student Researcher Aug 2022 – May 2023
- Developed theory for diffusion models, and wrote experiments to improve understanding, resulting in a paper at NeurIPS 2024.
- Developed mean estimation algorithms, resulting in multiple papers at NeurIPS, ICML and COLT.

Sigma Computing, Inc. *San Francisco, CA*
Research Intern May 2022 – Aug 2022
- Worked on designing and implementing anomaly detection algorithms for various datasets

University of Wisconsin, Madison *Madison, WI*
Research Intern May 2020 – Aug 2020
- Studied gradient descent algorithms to robustly estimate the mean of a high-dimensional Gaussian
- Wrote experiments for robust sparse estimation in Python. Resulted in NeurIPS 2022 paper.

Jane Street *New York, NY*
Software Developer Intern May 2016 – Aug 2016
- Wrote server code to employ state machine replication to send and receive data via RPCs
- Developed a market data parser in OCaml

Bloomberg *New York, NY*
R&D Intern May 2015 – Aug 2015
- Developed real-time system for Bloomberg Terminal to track messages between services in C++

PAPERS

14. **Posterior Sampling by Combining Diffusion Models with Annealed Langevin Dynamics**
Zhiyang Xun, Shivam Gupta, Eric Price
Neural Information Processing Systems (NeurIPS) 2025

13. **Efficient Knowledge Distillation via Curriculum Extraction**
Shivam Gupta, Sushrut Karmalkar
International Conference on Learning Representations (ICLR) 2025 DeLTa Workshop

12. **Uncovering the Bias in Diffusion-based Posterior Sampling Methods**
Shivam Gupta, Brett Levac, Jon Tamir, Eric Price
Manuscript

11. **Faster Diffusion Sampling with Randomized Midpoints: Sequential and Parallel**
    Shivam Gupta, Linda Cai, Sitan Chen
    International Conference on Learning Representations (ICLR) 2025
10. **Improved Sample Complexity Bounds for Diffusion Model Training**
    Shivam Gupta, Aditya Parulekar, Eric Price, Zhiyang Xun
    Neural Information Processing Systems (NeurIPS) 2024
9. **Diffusion Posterior Sampling is Computationally Intractable**
    Shivam Gupta, Ajil Jalal, Aditya Parulekar, Eric Price, Zhiyang Xun
    International Conference on Machine Learning (ICML) 2024
8. **Beyond Catoni: Sharper Rates for Heavy-Tailed and Robust Mean Estimation**
    Shivam Gupta, Samuel B. Hopkins, Eric Price
    Conference on Learning Theory (COLT) 2024
7. **Minimax-Optimal Location Estimation**
    Shivam Gupta, Jasper C.H. Lee, Eric Price, Paul Valiant
    Neural Information Processing Systems (NeurIPS) 2023
6. **Finite-Sample Symmetric Mean Estimation with Fisher Information Rate**
    Shivam Gupta, Jasper C.H. Lee, Eric Price
    Conference on Learning Theory (COLT) 2023
5. **High-dimensional Location Estimation via Norm Concentration for Subgamma Vectors**
    Shivam Gupta, Jasper C.H. Lee, Eric Price
    International Conference on Machine Learning (ICML) 2023
4. **Finite-Sample Maximum Likelihood Estimation of Location**
    Shivam Gupta, Jasper C.H. Lee, Eric Price, Paul Valiant
    Neural Information Processing Systems (NeurIPS) 2022
3. **Outlier-Robust Sparse Estimation via Non-Convex Optimization**
    Yu Cheng, Ilias Diakonikolas, Rong Ge, Shivam Gupta, Daniel Kane, Mahdi Soltanolkotabi
    Neural Information Processing Systems (NeurIPS) 2022
2. **Sharp Constants in Uniformity Testing via the Huber Statistic**
    Shivam Gupta, Eric Price
    Conference on Learning Theory (COLT) 2022
1. **Nash Equilibrium Computation in Resource Allocation Games**
    Shivam Gupta, Ruta Mehta
    International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2018

SELECTED AWARDS

- COLT 2024 Travel Award
- C.W. Gear Outstanding Undergraduate Student Award 2018
- Conference Travel Grant 2018 (for travel to AAMAS)
- Horace and Kate King Wu International Undergraduate Scholarship 2018
- Illinois Engineering Achievement Scholarship 2017
- NTT Data, Inc., Scholarship 2015, 2016
- Franz Hohn and J.P. Nash Scholarship 2015 (for research in scientific computing)
- Perfect score in the Indian National Olympiad in Informatics, and selected as one of 26 students to attend the International Olympiad in Informatics training camp 2014
- ACM ICPC Mid-Central Regional: Team placed 6th in 2017, 4th in 2016
- Represented India in SEARCC Software Competition 2013, Colombo, Sri Lanka, and placed 3rd
- Placed 2nd in Dropbox Open programming contest 2015 at UIUC
- Won Bloomberg CodeCon Challenge and invited to CppCon 2015 in Bellevue, Washington
- Won 3Red Trading Tech Challenge in 2015 and 2016 and invited to Chicago

OTHER EXPERIENCE

Research with Prof. Andreas Klockner (UIUC) Aug 2014 –May 2017

- Developed and implemented mesh-refinement algorithms (in Python and using NumPy) to iteratively refine and coarsen meshes while preserving connectivity information.

SKILLS

- *Languages:* Python, C/C++, Java, OCaml, Haskell, JavaScript
- *Software and Libraries:* PyTorch, JAX, TensorFlow, NumPy, SciPy, Mathematica, LaTeX

SERVICE

- *Program Committee:* ALT 2025, 2026; AAAI 2025, 2026
- *Reviewing:* SODA 2021, 2024; NeurIPS 2023, 2024, 2025, 2026; ITCS 2024; ALT 2024, 2025, 2026; ICLR 2024, 2025, 2026; ICML 2024, 2025, 2026; TF2M@ICML 2024; AISTATS 2025, 2026; FOCS 2025; MOSS@ICML 2025;

TALKS

- **Efficient Knowledge Distillation via Curriculum Extraction** April 2025
  Google DeepMind
- **Diffusion Models: Sampling, Training and Reconstruction** November 2024
  Boson AI
- **Theoretical Insights for Practical Generative Modeling** November 2024
  Google Research
- **Diffusion Models: Sampling, Training and Reconstruction** October 2024
  Stability AI
- **Beyond Catoni: Sharper Rates for Heavy-Tailed and Robust Mean Estimation** July 2024
  Conference on Learning Theory 2024 (Online)
- **Sample-Efficient Training for Diffusion** November 2023
  IFML Workshop on Generative AI, UT Austin
- **A Finite-Sample Theory for Mean Estimation with Fisher Information Rate** October 2023
  MIT Algorithms and Complexity Seminar
- **A Finite-Sample Theory for Mean Estimation with Fisher Information Rate** October 2023
  CMU Theory Lunch
- **Finite-Sample Symmetric Mean Estimation with Fisher Information Rate** July 2023
  Conference on Learning Theory 2023, Bangalore, India
- **High-dimensional Estimation via Norm Concentration for Subgamma Vectors** July 2023
  International Conference on Machine Learning 2023 (Online)
- **Sharp Constants in Uniformity Testing via the Huber Statistic** July 2022
  ML Tea, UT Austin

TEACHING

At UT Austin:

- *Teaching Assistant*, Sublinear Algorithms (CS 395T) Fall 2020
- *Teaching Assistant*, Introduction to Algorithms (CS 331) Spring 2019
- *Teaching Assistant*, Machine Learning (CS 395T) Summer 2019, Fall 2019
- *Teaching Assistant*, Elements of Data Analytics (CS 329E) Spring 2019
- *Teaching Assistant*, Elements of Computer Programming (CS 303E) Fall 2018

At UIUC:

- *Course Assistant*, Introduction to Algorithms (CS 374) Spring 2018
- *Grader*, Algorithms II (CS 473) Spring 2018

RELEVANT COURSES

Probability and Stochastic Processes, Learning Theory, Coding Theory, Theoretical Statistics, Randomized Algorithms, Markov Chains and Mixing Time, Approximation Algorithms, Combinatorial Mathematics, Numerical Linear Algebra, Wireless Networking